Wearable Biomarkers For Parkinson’s Disease

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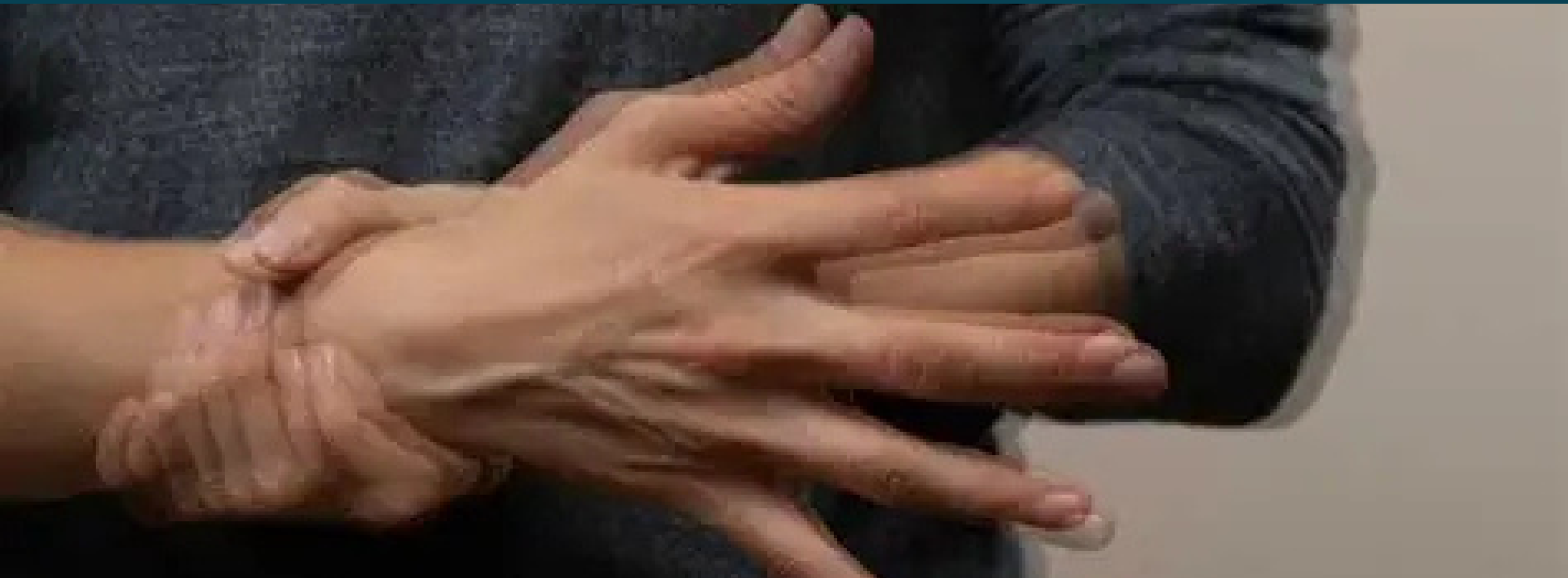
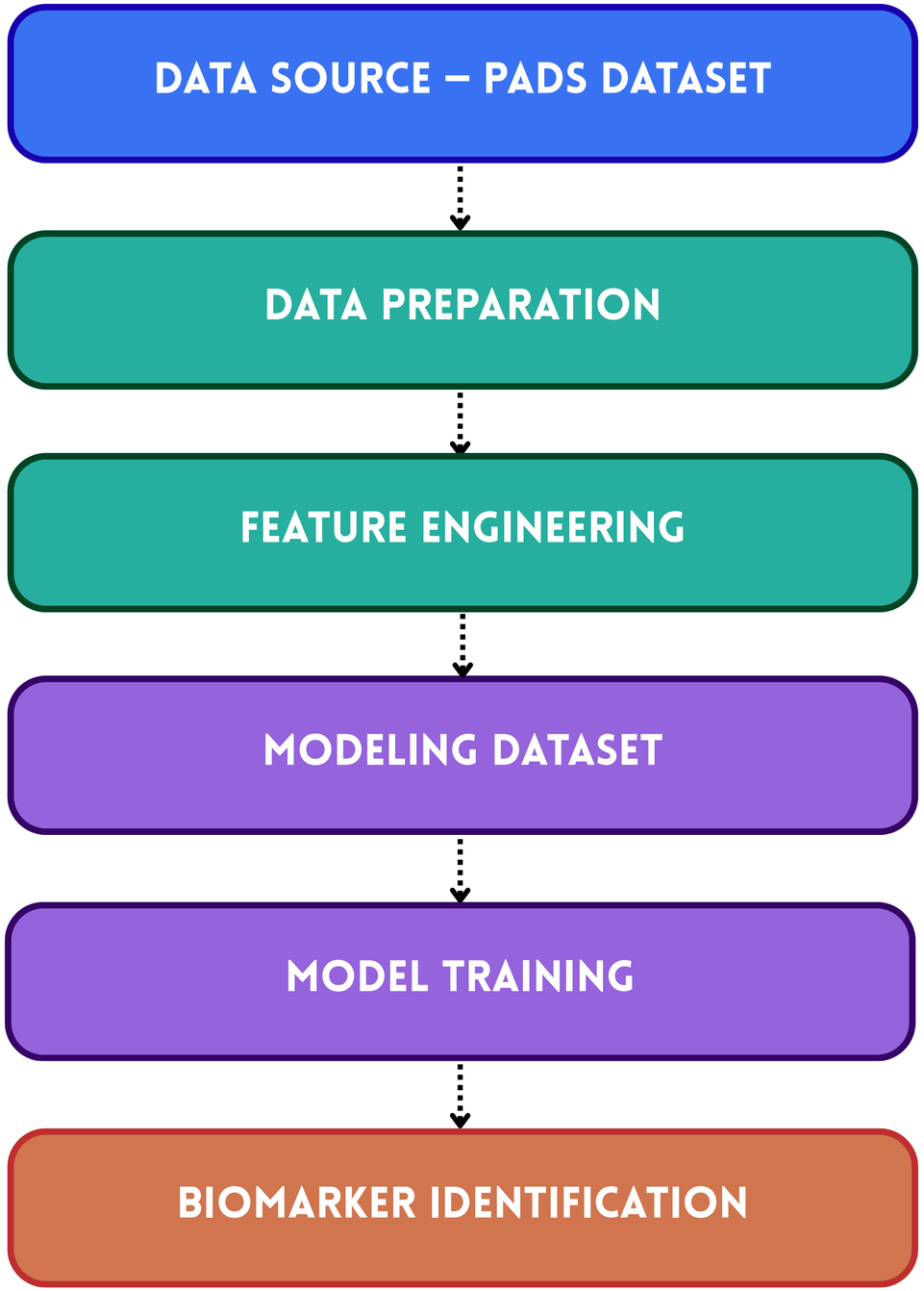
Behavioral Health Informatics



To what extent can smartwatch-based sensor features complement or replicate the predictive power of self-reported questionnaire data for early detection of Parkinson's disease, and how can the combination of both improve predictive performance?

Description: Using the Parkinson's Disease Smartwatch Dataset (PADS) from PhysioNet, which includes smartwatch motion signals and clinical questionnaires from 469 participants, we apply a Machine Learning model to classify PD vs non-PD subjects.

Goal: The project focuses on identifying the most predictive features derived from wearable sensors, providing insights into potential digital biomarkers for Parkinson's screening.



DATA PREPARATION

QUESTIONNAIRE

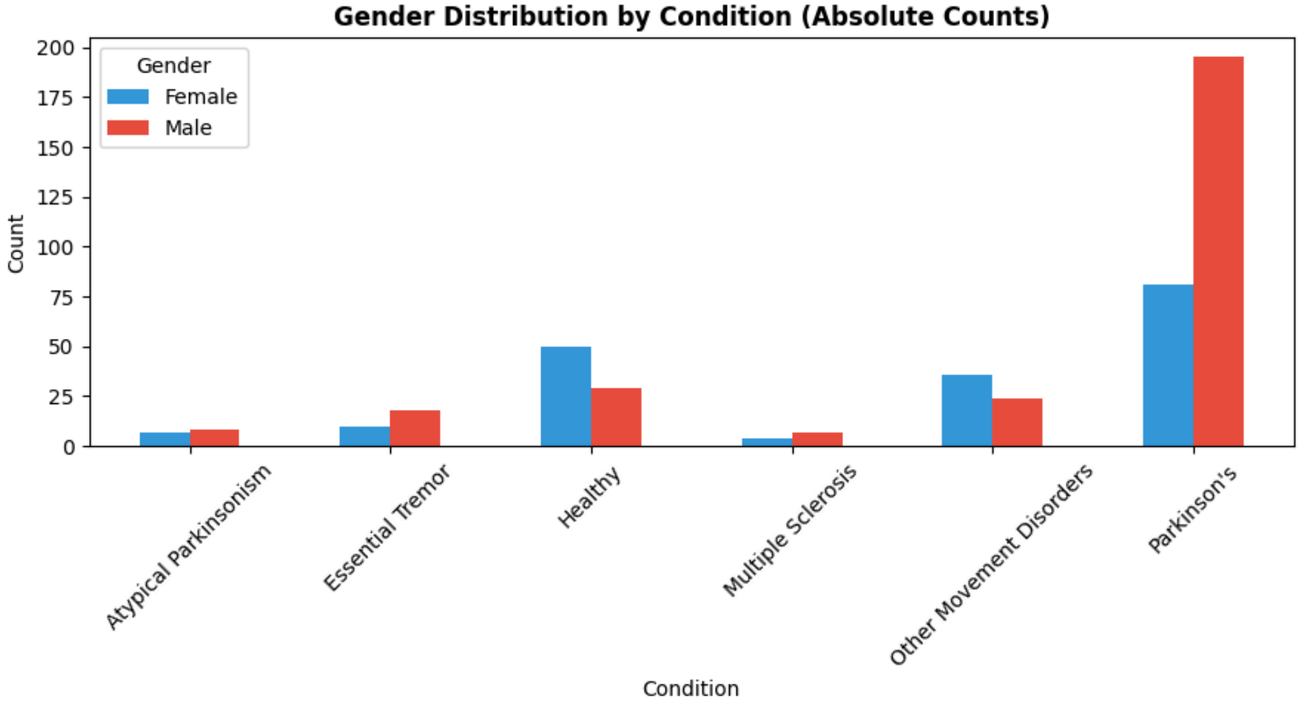
- Boolean responses → numeric variables
- Integration with clinical annotations
- Data quality checks:
  - Validation of data types
  - Handling of missing values
  - Detection of duplicates

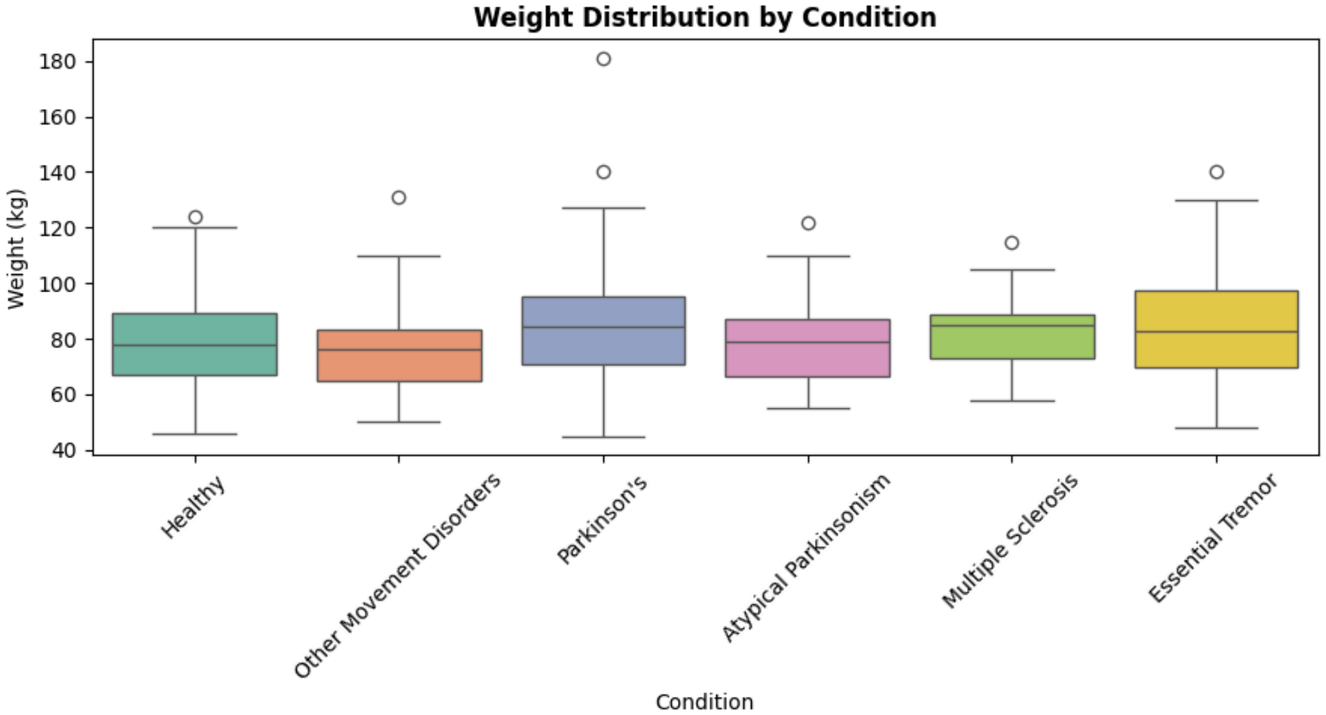
MOVEMENTS

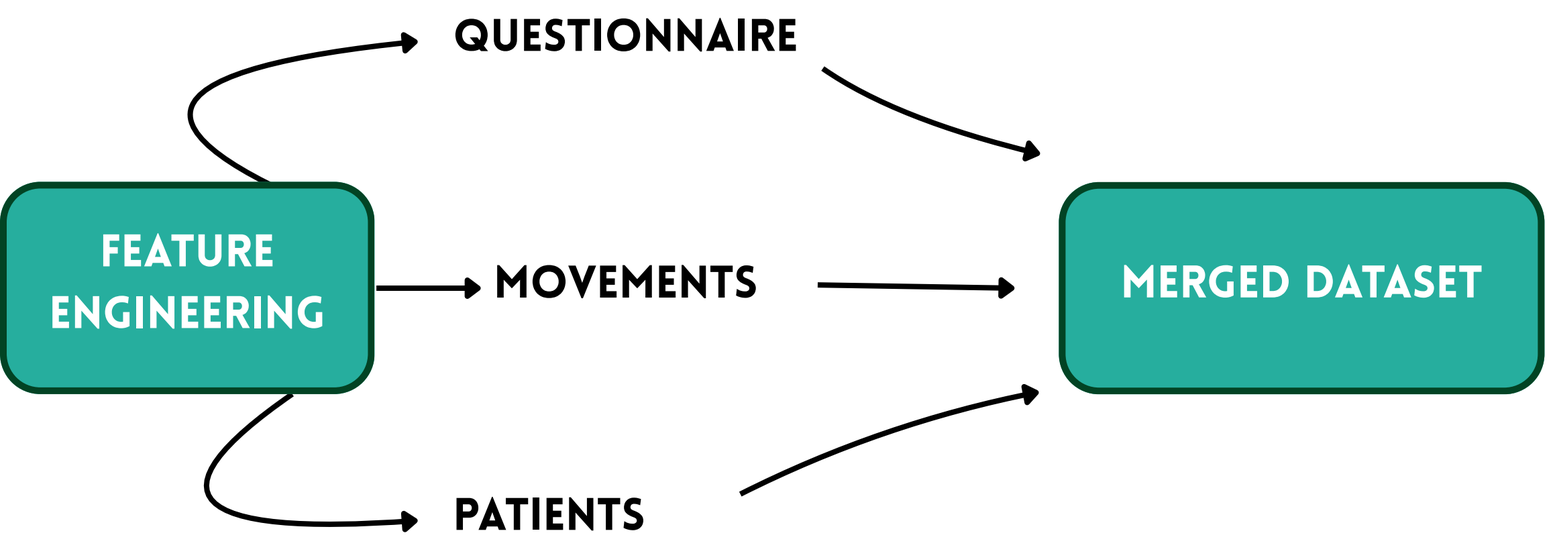
- Linking sensor files to subjects and clinical conditions
- Extraction of features from time-series data (saving in a csv file)
- EDA:
  - Assessment of missing data
  - Summary statistics by condition
  - Visualization via boxplot
  - Outlier detection using the IQR method

PATIENTS

- Unified patient table from JSON records
- Descriptive statistics (age, height, weight, ...)
- Analysis of target distribution by clinical condition
- Data quality checks:
  - Validation of data types
  - Handling of missing values
  - Detection of duplicates

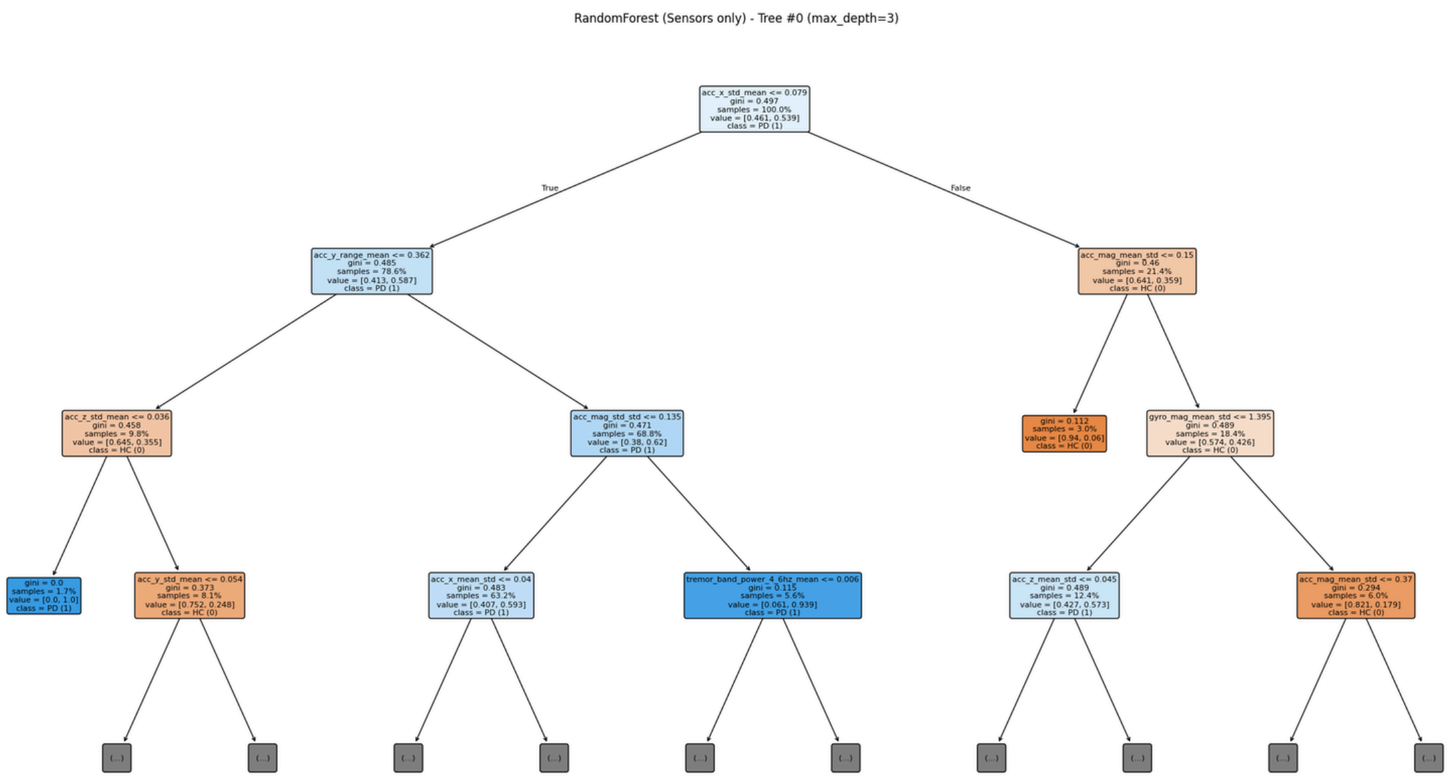

Gender Distribution by Condition (Absolute Counts)
Gender
Female
Male
Count
0
25
50
75
100
125
150
175
200
Atypical Parkinsonism
Essential Tremor
Healthy
Multiple Sclerosis
Other Movement Disorders
Parkinson's
Condition


Weight Distribution by Condition
Weight (kg)
40
60
80
100
120
140
160
180
Healthy
Other Movement Disorders
Parkinson's
Atypical Parkinsonism
Multiple Sclerosis
Essential Tremor
Condition

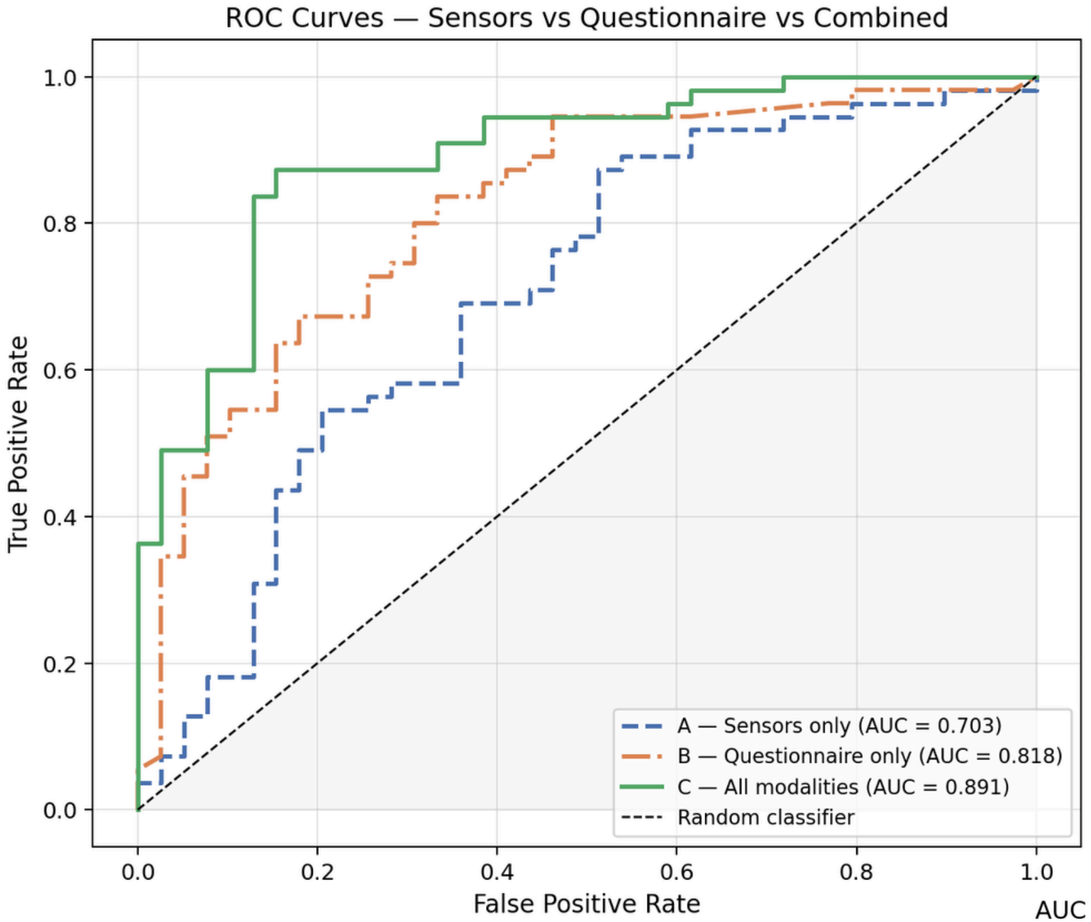


- Load csv Data
- Merge file CSV (patient_df)
  - Sensor Feature Aggregation
- Age-aware stratified split
- Cutting demographic data
- Developing 3 set of data:
  - Sensor only
  - Questionnaire only
  - Combined (12+13)
- Save train and test dataset

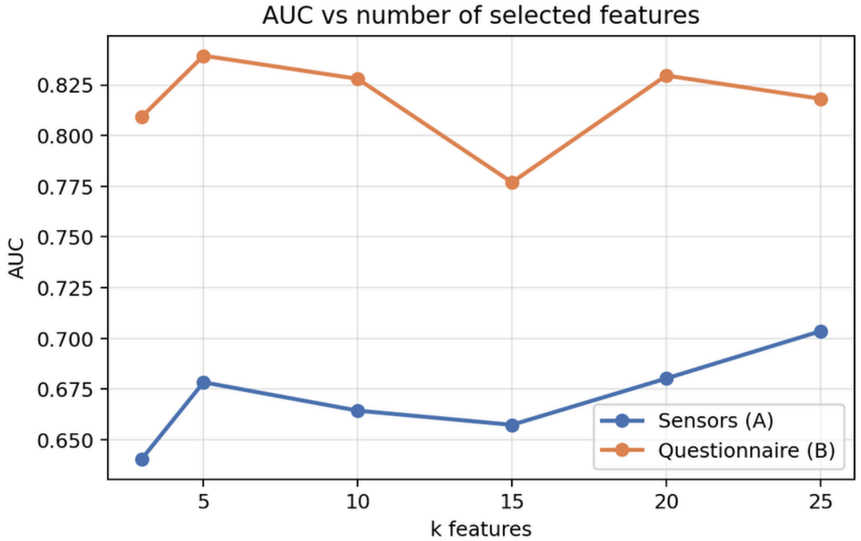
MODEL TRAINING

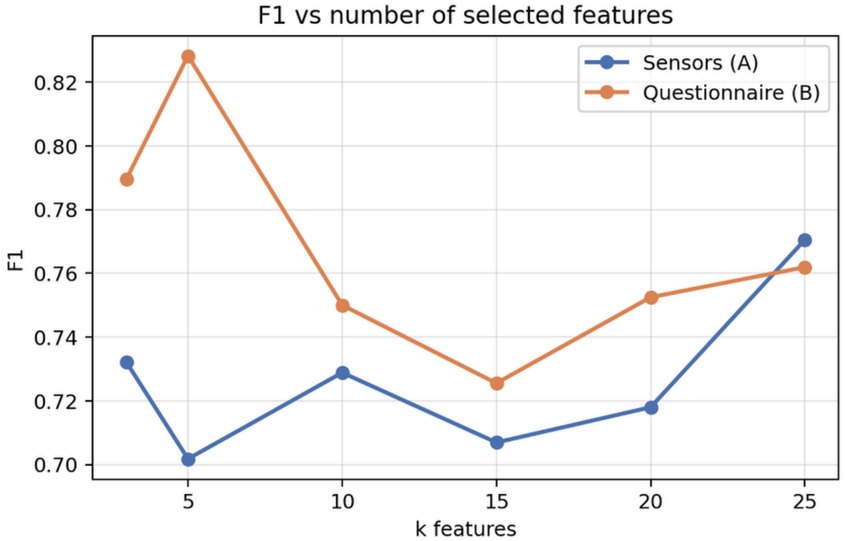
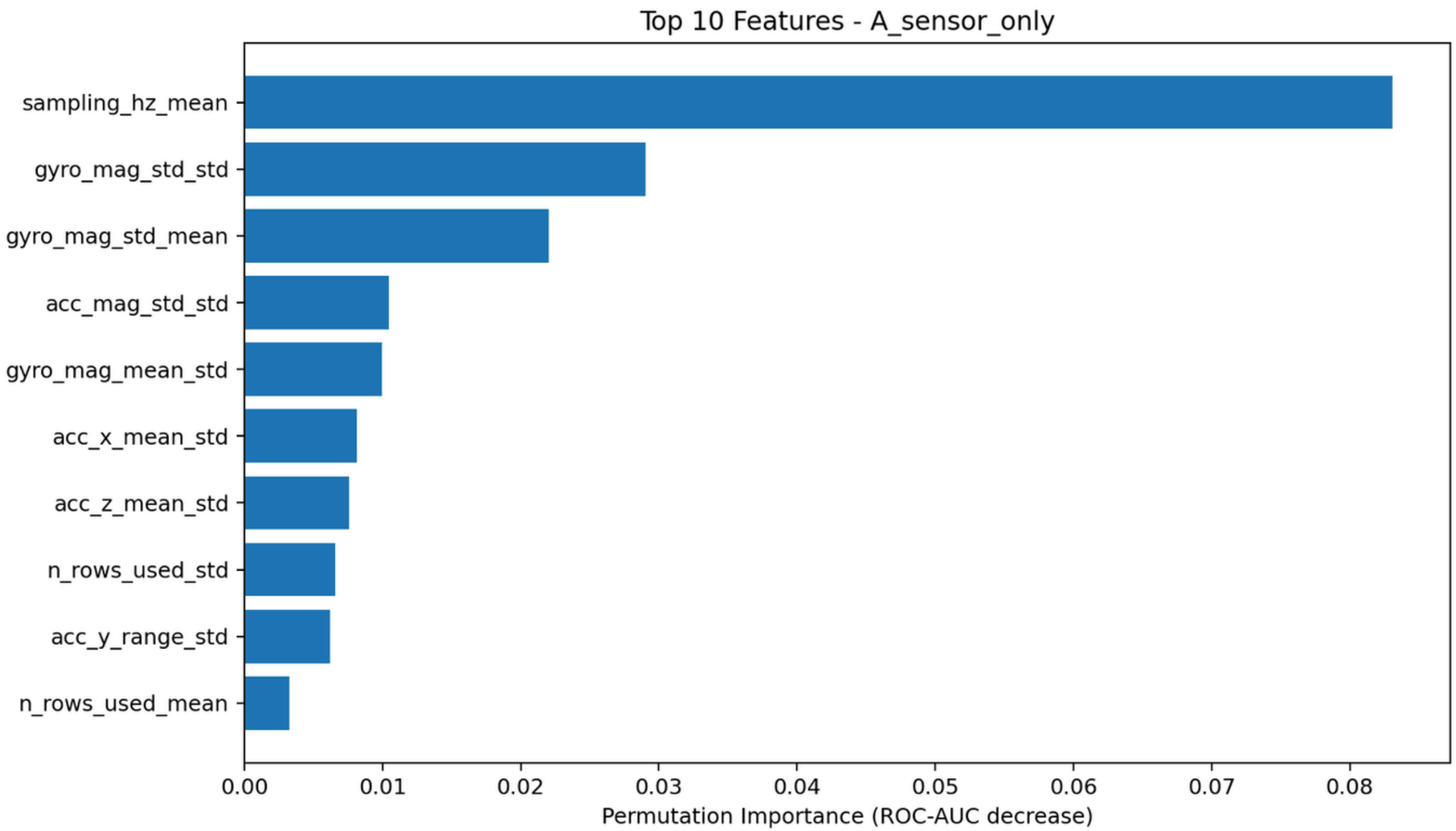


- Data preparation
- Hyperparameters tuning:
  - Number of trees
  - Tree depth
  - Split / minimum leaf size
  - Number of features considered at each split
  - Optional class balancing
- Training
- Evaluation


ROC Curves — Sensors vs Questionnaire vs Combined
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
A — Sensors only (AUC = 0.703)
B — Questionnaire only (AUC = 0.818)
C — All modalities (AUC = 0.891)
Random classifier

RESULTS


AUC vs number of selected features
AUC
k features
0.650
0.675
0.700
0.725
0.750
0.775
0.800
0.825
5
10
15
20
25
Sensors (A)
Questionnaire (B)


F1 vs number of selected features
F1
k features
0.70
0.72
0.74
0.76
0.78
0.80
0.82
5
10
15
20
25
Sensors (A)
Questionnaire (B)

BIOMARKER IDENTIFICATION